

(a) textbook hard-threshold ABC-PMC: one update per generation

P_{t+1} : the N kept, weighted; the rejected are discarded

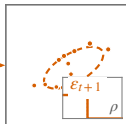
barrier: q_{t+1} needs the complete P_{t+1}

the current population only; earlier generations are discarded



1 state

next threshold: a quantile of the population's discrepancies; accept or reject



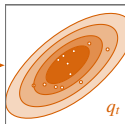
2 tolerance ϵ

weights carried over from the last generation; covariance from them



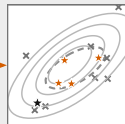
3 weights, Σ

one mixture proposal over the whole population



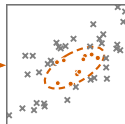
4 proposal

propose, simulate, accept or reject, one candidate at a time, until N are kept



5 draw

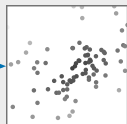
the kept are weighted against that one proposal, after acceptance



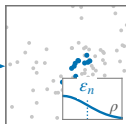
6 weight

(b) generation-free (ours): one update per arrival

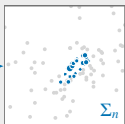
$|\mathcal{H}_n| < k$: prior draw, $w^* = 1$, no ϵ stamp



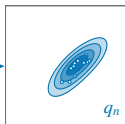
every particle ever evaluated, with its stored bandwidth and weight



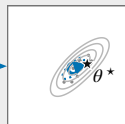
the k best form the archive; a smooth kernel of width ϵ_n , which only ever tightens



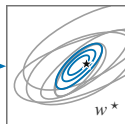
archive weights: stored weight times kernel weight; covariance from them



one mixture proposal over the archive, rebuilt at every call



one candidate: a parent drawn by weight, then perturbed



weighted against the mixture of all proposals used so far, before it is simulated

θ^* to a free worker; ρ^* arrives later, among other workers' results: append $(\theta^*, \rho^*, \epsilon_{\tau^*}, \tau^*, w^*), \tau^* = n$